

Q3: Ethical Auditing & Documentation Debt Mitigation

Sound Check Audit · Privacy-Preserving AI · Fair ASR Training

Dataset	Mozilla Common Voice (demo: synthetic 5,000 utterances)
Tools	Python 3.10 · PyTorch 2.x · Hugging Face Transformers
Evaluation	FAD (Fréchet Audio Distance) · DNSMOS proxy · STOI proxy
Deliverables	audit.py · privacymodule.py · pp_demo.py · train_fair.py · evaluation_scripts/ · audit_plots.pdf

1 · Bias Identification & Documentation Debt Audit

1.1 Dataset Overview

We audited Mozilla Common Voice English (simulated via 5,000 synthetic utterances mirroring real Common Voice statistics). Three demographic axes were examined: **Gender**, **Age Group**, and **Accent/Dialect**.

1.2 Documentation Debt

Field	Missing %	Assessment
Gender	5.7%	Acceptable – minor gap
Age	11.8%	Moderate – targeted re-labelling recommended
Accent	14.5%	High – significant documentation debt

1.3 Representation Bias Results

Metric	Gender	Age	Accent
Normalised Entropy	0.611	0.914	0.886
Gini Coefficient	0.542	0.319	0.357
χ^2 p-value	<0.001	<0.001	<0.001
Verdict	BIASED	BIASED	BIASED

1.4 Key Findings

- Gender: Male speakers (68.7%) dominate; female (23.4%) and other/non-binary voices are severely underrepresented. Normalised entropy of 0.611 vs ideal 1.0 indicates strong skew.

- Age: Speakers in their 30s–40s dominate (46.7% combined); teens, elderly (60+), and children have <10% combined representation. Models trained on this data will likely degrade for these populations.
- Accent/Dialect: US accent (37.7%) and British English (17.9%) dominate. Non-Western accents (Indian, African, Asian) are critically underrepresented despite being spoken by billions of people globally.
- Intersectional Gap: The male × thirties cell alone occupies 19.1% of total data. Female × elderly combinations each represent <0.5%.

2 · Privacy-Preserving Voice Transformation

2.1 Architecture

The **PrivacyModule** (privacymodule.py) implements a CycleGAN-inspired voice conversion system with disentangled representations:

- **ContentEncoder**: Strided 1D convolutions + residual blocks strip speaker identity, retaining only phonetic content (z_content).
- **AttributeEncoder**: Global average pooling over a parallel conv branch produces a speaker attribute vector (z_attr).
- **TargetAttributeEmbedder**: Maps discrete gender/age labels to a learned embedding space, allowing arbitrary target attribute specification.
- **Decoder**: AdaIN-based residual decoder injects target style into content to generate the transformed mel spectrogram.
- **Vocoder**: Griffin-Lim inversion (training mode) or learned HiFi-GAN (production). Outputs raw waveform.

2.2 Training Objectives

Four losses are combined to balance transformation quality and content fidelity:

Loss Term	Formula	Weight	Purpose
Reconstruction	$L1(\text{mel_recon}, \text{mel_src})$	$\lambda=10$	Cycle: $\text{src} \rightarrow \text{tgt} \rightarrow \text{src}$
Content-Cycle	$\text{MSE}(z_{\text{cont_cyc}}, z_{\text{cont}})$	$\lambda=5$	Preserve phonemes
Adversarial	$\text{MSE}(D(\text{mel_fake}), 1)$	$\lambda=1$	Realism
Patch Discrim.	$\text{MSE}(D(\text{real}), 1) + \text{MSE}(D(\text{fake}), 0)$	$\times 0.5$	GAN stability

2.3 Supported Transformations

The module supports any combination of supported gender and age attributes. Example transformation pairs (demonstrated in pp_demo.py):

- male/old → female/young (primary use case)
- female/young → male/old
- male/old → female/old (gender-only)
- male/young → female/middle

Note: With random weight initialisation (no training), transformations are structurally valid but acoustically random. Real-world quality requires training on ≥ 50 hours of multi-speaker data per demographic group.

3 · Fairness Loss & Fair ASR Training

3.1 Model Architecture

The **FairASRModel** (train_fair.py) is a CTC-based ASR system consisting of a strided convolutional feature extractor (5 conv layers, total stride $\times 80$) followed by 6 Conformer blocks (d=256, 4 heads) and a linear CTC projection.

3.2 Fairness Loss Design

Standard CTC training minimises the mean loss across all utterances, which allows the model to perform well on majority groups while failing silently on minorities. Our **FairnessLoss** adds three terms:

$$L_{\text{total}} = L_{\text{CTC}} + \lambda_{\text{fair}} \cdot (\text{worst_g} - \text{best_g}) + \lambda_{\text{reg}} \cdot \sum_g \max(0, L_g - \tau)^2$$

- L_{CTC} : Standard mean CTC loss across the batch.
- Gap penalty ($\lambda_{\text{fair}}=0.5$): Penalises the difference between the worst-performing and best-performing group in each batch. Forces the model to improve weak groups.
- Threshold regulariser ($\lambda_{\text{reg}}=0.1$, $\tau=0.05$): Quadratic penalty for any group whose loss exceeds the acceptable threshold τ .
- Group-DRO EMA weights: Exponential moving averages track per-group losses across batches to provide stable gradient signals even when groups are sparsely sampled.

3.3 Demographic Groups

Groups are defined by the cross-product of coarse gender $\times$ age buckets derived from the audit:

Group	Description	% in Demo Data
male_young	Male, teens–twenties	~18%
male_middle	Male, thirties–forties	~33%
male_old	Male, fifties+	~12%
female_young	Female, teens–twenties	~6%
female_middle	Female, thirties–forties	~10%
female_old	Female, fifties+	~3%
other	Non-binary / unlabelled gender	~18%

4 · Validation: FAD & DNSMOS Evaluation

4.1 Fréchet Audio Distance (FAD)

FAD measures the distributional distance between reference audio and privacy-transformed audio in the embedding space of a VGGish-style neural network (128-dim proxy embeddings). It generalises FID to audio.

Scenario	FAD	Interpretation
Identical distribution	~0.0	Perfect (baseline)
Small shift (high quality)	~2-5	Good – minor artefacts
Medium shift	~10-25	Acceptable
Large shift / mode collapse	100+	Fail – retrain needed
Our privacy transform	<15	Target (no toxicity traps)

4.2 DNSMOS Proxy

DNSMOS estimates MOS (Mean Opinion Score, 1–5) non-intrusively using a GRU network on STFT magnitude features. Three sub-scores are reported:

- **SIG** (Signal Quality): Measures speech clarity and absence of signal-level distortion. Target: ≥ 3.5 .
- **BAK** (Background): Measures background noise level and absence of artifacts introduced by the transformation. Target: ≥ 3.5 .
- **OVR** (Overall MOS): Weighted combination of SIG and BAK. Target: ≥ 3.0 to avoid 'Toxicity Traps'.

A 'Toxicity Trap' is defined as a transformation that passes FAD (<30) but introduces linguistic distortions or acoustic artefacts that degrade ASR accuracy or listener experience. Both FAD and DNSMOS are necessary – FAD alone can miss localised artefacts while DNSMOS alone can miss distributional drift.

4.3 STOI Proxy (Intelligibility)

Short-Time Objective Intelligibility (STOI) is computed as the average frame-level correlation between reference and transformed magnitude spectra. Values near 1.0 indicate preserved intelligibility. We require $\text{STOI} > 0.85$ for the transformation to be considered linguistically safe.

5 · Ethical Considerations

5.1 Dual-Use Risk

Voice transformation technology can be misused for: deepfakes, impersonation, non-consensual identity manipulation, and evidence fabrication. Our implementation includes:

- Watermarking: Production deployments should embed imperceptible watermarks in transformed audio (not implemented in this prototype).
- Consent framework: Transformations should only be applied with explicit speaker consent for research/anonymisation purposes.
- Audit trails: All transformation operations should be logged with purpose, source attributes, and authorisation.

5.2 Fairness Loss Limitations

- Group definition risk: Coarse gender/age buckets may reinforce binary categorisation. Future work should use continuous embeddings.
- Proxy labels: Using self-reported demographic labels (as in Common Voice) introduces noise and may not reflect perceived demographic attributes.
- Intersectional blindness: The current fairness loss operates on single-axis groups; intersectional groups (e.g. female × old × non-native accent) may still be overlooked without explicit multi-axis group definitions.
- Fairness–accuracy trade-off: A fairness penalty of $\lambda=0.5$ may reduce overall WER slightly (<5% relative) while closing the worst-group gap. This trade-off should be explicitly reported in production deployments.

5.3 Recommendations

- Re-collect data targeting under-represented groups (female, elderly, non-English-accented speakers) with community partnerships.
- Adopt participatory data governance: involve speaker communities in deciding how their voices are used and transformed.
- Publish fairness evaluation results alongside WER metrics in all model cards and technical reports.
- Adopt continuous re-auditing pipelines; run audit.py on each new dataset release as part of CI/CD.

Generated by q3/ submission pipeline. All code is original and implemented using Python, PyTorch, and Hugging Face. Dataset: Mozilla Common Voice (demo mode). Tools: audit.py · privacymodule.py · pp_demo.py · train_fair.py · evaluation_scripts/fad_eval.py · evaluation_scripts/dnsmos_eval.py